Homework 10

This homework is due at 11 PM on April 10, 2026.

Submission Format: Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned), as well as a printout of your completed Jupyter notebook(s).

1. **Sensitivity and Dual Variables**

In this problem, we explore the interpretation of dual variables as sensitivity parameters of the primal problem. Recall the canonical convex primal problem

$$\min_{\vec{x}\in\mathbb{R}^n} \quad f_0(\vec{x}) \tag{1}$$

$$\text{s.t.} \quad f_i(\vec{x}) \leq 0, \quad i = 1, \ldots, m \tag{2}$$

$$h_j(\vec{x}) = 0, \quad j = 1, \ldots, p \tag{3}$$

where f_i are convex for all $i = 0, \ldots, m$ and h_j are affine for all $j = 1, \ldots, p$.

For $\vec{u} = (u_1, \ldots, u_m)^\top \in \mathbb{R}^m$ and $\vec{v} = (v_1, \ldots, v_p)^\top \in \mathbb{R}^p$, we consider the *perturbed* problem

$$p^\star(\vec{u}, \vec{v}) = \min_{\vec{x}\in\mathbb{R}^n} \quad f_0(\vec{x}) \tag{4}$$

$$\text{s.t.} \quad f_i(\vec{x}) \leq u_i, \quad i = 1, \ldots, m \tag{5}$$

$$h_j(\vec{x}) = v_j, \quad j = 1, \ldots, p \tag{6}$$

In other words, $p^\star(\vec{u}, \vec{v})$ is a function of $\vec{u}$ and $\vec{v}$ that gives the optimal value for the perturbed problem (if it is feasible). If the problem is infeasible (i.e. no points exist that satisfy the constraints), we say that $p^\star(\vec{u}, \vec{v}) = +\infty$. Note that $p^\star(\vec{0}, \vec{0})$ is the value of the original problem.

(a) Prove that $p^\star(\vec{u}, \vec{v})$ is jointly convex[1] in $(\vec{u} \in \mathbb{R}^m,\ \vec{v} \in \mathbb{R}^p)$.

HINT: Let

$$\mathcal{D} \doteq \{(\vec{x} \in \mathbb{R}^n, \vec{u} \in \mathbb{R}^m, \vec{v} \in \mathbb{R}^p) \mid f_i(\vec{x}) \leq u_i \quad \forall i, \quad h_j(\vec{x}) = v_j \quad \forall j\}, \tag{7}$$

which is the set of all triples $(\vec{x}, \vec{u}, \vec{v})$ such that $\vec{x}$ is a feasible point for the perturbed problem with the perturbations $(\vec{u}, \vec{v})$. Show that $\mathcal{D}$ is convex. Now define $F(\vec{x}, \vec{u}, \vec{v})$ to be a function that is equal to $f_0(\vec{x})$ on $\mathcal{D}$ and $+\infty$ otherwise. Prove that $F(\vec{x}, \vec{u}, \vec{v})$ is jointly convex in $(\vec{x}, \vec{u}, \vec{v})$, and then observe that

$$p^\star(\vec{u}, \vec{v}) = \min_{\vec{x}} F(\vec{x}, \vec{u}, \vec{v}). \tag{8}$$

From here, to show that $p^\star(\vec{u}, \vec{v})$ is jointly convex in $(\vec{u}, \vec{v})$, you may prove and use the following lemma: Let S_1, S_2 be convex sets with a function $f\colon S_1 \times S_2 \to \mathbb{R}$ which is jointly convex in both arguments. Define $g(\vec{y}) = \min_{\vec{x}\in S_1} f(\vec{x}, \vec{y})$. Then $g(\vec{y})$ is convex in $\vec{y} \in S_2$.

(b) Assume that strong duality holds, and that the dual optimum is attained, for the unperturbed primal problem (1). Let $(\vec{\lambda}^\star, \vec{\sigma}^\star)$ be the optimal dual variables for the dual of (1). Show that for any point $\vec{z}$ that is feasible for the perturbed problem (4), we have

$$f_0(\vec{z}) \geq p^\star(\vec{0}, \vec{0}) - \vec{u}^\top \vec{\lambda}^\star - \vec{v}^\top \vec{\sigma}^\star \tag{9}$$

[1] Recall that a function $f\colon A \times B \to \mathbb{R}$ is jointly convex in $(\vec{a} \in A,\ \vec{b} \in B)$ if $A \times B$ is convex, and for all $\theta \in [0, 1]$, and for all $\vec{a}_1, \vec{a}_2 \in A,\ \vec{b}_1, \vec{b}_2 \in B$, we have that $f(\theta\vec{a}_1 + (1-\theta)\vec{a}_2,\ \theta\vec{b}_1 + (1-\theta)\vec{b}_2) \leq \theta f(\vec{a}_1, \vec{b}_1) + (1-\theta) f(\vec{a}_2, \vec{b}_2)$.

HINT: Let L and g be the Lagrangian and dual function, respectively, of the unperturbed primal problem (1). Use strong duality to relate $p^\star(\vec{0}, \vec{0})$ to $g(\vec{\lambda}^\star, \vec{\sigma}^\star)$. Upper-bound the value of $g(\vec{\lambda}^\star, \vec{\sigma}^\star)$ by the value of $L(\vec{z}, \vec{\lambda}^\star, \vec{\sigma}^\star)$. Then, noting that $\vec{z}$ is feasible for the perturbed problem (4), apply bounds on $f_i(\vec{z})$ and $h_j(\vec{z})$ to bound $L(\vec{z}, \vec{\lambda}^\star, \vec{\sigma}^\star)$.

(c) Using the result of part 1(b), show that for all $\vec{u}, \vec{v}$, we have

$$p^\star(\vec{u}, \vec{v}) \geq p^\star(\vec{0}, \vec{0}) - \vec{u}^\top \vec{\lambda}^\star - \vec{v}^\top \vec{\sigma}^\star. \tag{10}$$

(d) Suppose we only have 1 equality and 1 inequality constraint (that is u, v are scalars). For each (u, v), exactly one of the following three cases must apply: either we must have $p^\star(u, v) > p^\star(0, 0)$, we must have $p^\star(u, v) < p^\star(0, 0)$, or it is impossible to conclude which one is greater. For each of the following situations, assume that $|u| \approx |v| \approx 1$, the small Lagrange multiplier has absolute value ≈ 1, the large Lagrange multiplier has absolute value ≈ 10, and argue which case applies.

i. $\lambda^\star$ is large (as compared with $\sigma^\star$) and $u < 0$.

ii. $\lambda^\star$ is large (as compared with $\sigma^\star$) and $u > 0$.

iii. $\sigma^\star$ is large (as compared with $\lambda^\star$) and positive and $v < 0$.

iv. $\sigma^\star$ is large (as compared with $\lambda^\star$) and negative and $v > 0$.

HINT: Use the bound you computed in part 1(c).

Note that we can think of u and v as variables we choose — by examining how the solution to our original primal problem changes, we can describe how "sensitive" our problem is to its different constraints!

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2. KKT with Circles

Consider the problem

$$\min_{\vec{x}\in\mathbb{R}^2} \quad x_1^2 + x_2^2 \tag{11}$$

$$\text{s.t.} \quad (x_1-1)^2 + (x_2-1)^2 \leq 2 \tag{12}$$

$$(x_1-1)^2 + (x_2+1)^2 \leq 2 \tag{13}$$

where $x = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^\top \in \mathbb{R}^2$.

(a) Sketch the feasible region and the level sets of the objective function. Find the optimal point $\vec{x}^*$ and the optimal value p^*.

(b) Does strong duality hold?

(c) Write the KKT conditions for this optimization problem. Do there exist Lagrange multipliers λ_1^* and λ_2^* that prove the optimality of $\vec{x}^*$?

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3. Strong Duality but No KKT

In this question, we will see an example of a problem where strong duality holds but the KKT conditions don't hold. Consider the following problem:

$$p^* = \min_{x_1, x_2 \in \mathbb{R}} \quad x_1^2 + x_2^2 \tag{14}$$

$$\text{s.t.} \quad (x_1 - 1)^2 + (x_2 - 1)^2 \leq 1 \tag{15}$$

$$(x_1 - 1)^2 + (x_2 + 1)^2 \leq 1. \tag{16}$$

(a) Sketch the feasible set. Find the optimal solution x^*.

(b) Write the KKT conditions and solve them.

(c) Formulate and solve the Lagrange dual problem. Does strong duality hold?

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4. Does Strong Duality Hold?

Consider

$$\min_{(x,y)\in\mathcal{D}} \quad \mathrm{e}^{-x} \tag{17}$$

$$\text{s.t.} \quad x^2/y \leq 0 \tag{18}$$

where $\mathcal{D} = \{(x, y) \mid y > 0\}$.

(a) Prove the problem is convex. Find the optimal value. *HINT: To prove the constraint function is convex, you will have to prove it is convex with respect to the vector* $\begin{bmatrix} x & y \end{bmatrix}^\top$. *Consider computing the Hessian of the constraint function, its determinant and trace, and show that it is PSD by analyzing signs of its eigenvalues.*

(b) Next, we will proceed to find an optimal solution and an optimal value for the dual problem. The Lagrangian dual function $g(\lambda)$, can be written as:

$$g(\lambda) = \inf_{(x,y)\in\mathcal{D}} \mathrm{e}^{-x} + \lambda\frac{x^2}{y}. \tag{19}$$

Explain why $g(\lambda)$ is lower bounded by 0 for $\lambda \geq 0$. *NOTE*: Here we are not dualizing the constraint $y > 0$ that is in the definition of $\mathcal{D}$ — this is only dualizing the other constraint.

(c) Show that $g(\lambda) = 0$ for $\lambda \geq 0$. *HINT: To show that the infimum in* (19) *is 0, we want to show there exist* (x, y) *such that both* e^{-x} *and* $\lambda\frac{x^2}{y}$ *can get arbitrarily close to 0. HINT: Consider a sequence* $\{x_k\}$ *going to* $+\infty$ *and a sequence* $\{y_k\}$ *also going to* $+\infty$ *such that* $\lim_{k\to\infty} \frac{x_k^2}{y_k} = 0$. *Simply put, we want to drive* x *to infinity in order to drive* e^{-x} *to* 0*, while having* y *grow faster than* x^2*, so that the second term also goes to* 0.

(d) Now, write the dual problem and find an optimal solution λ^* and an optimal value d^* for the dual problem using the results above. What is the duality gap?

(e) Does Slater's Condition hold for this problem? Does Strong Duality hold?

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5. Water Filling

Consider the following problem:

$$\text{minimize} \quad -\sum_{i=1}^{n} \log(\alpha_i + x_i) \tag{20}$$

$$\text{subject to} \quad \vec{x} \geq 0, \ \vec{1}^\top \vec{x} = c, \tag{21}$$

where $\alpha_i > 0$ for each $i = 1, \ldots, n$.

This problem arises in information theory, in allocating power to a set of n communication channels. The variable x_i represents the transmitter power allocated to the ith channel, and $\log(\alpha_i + x_i)$ gives the capacity or communication rate of the channel, so the problem is to allocate a total power of c to the channels, in order to maximize the total communication rate.

(a) Verify that this is a convex optimization problem with differentiable objective and constraint functions. Find the domain $\mathcal{D}$ of the objective function $-\sum_{i=1}^{n} \log(\alpha_i + x_i)$ where it is well defined.

(b) Let $\vec{\lambda} \in \mathbb{R}^n$ and $\nu \in \mathbb{R}$ be the dual variables corresponding to the constraints $x_i \geq 0, i = 1, \ldots, n$ and $\vec{1}^\top \vec{x} = c$, respectively. Write a Lagrangian for the optimization problem based on these dual variables.

(c) Write the KKT conditions for the problem.

(d) Since our problem is a convex optimization problem with differentiable objective and constraint functions, the KKT conditions provide sufficient conditions for optimality. Hence, we know that if we can find $\vec{x}^*$ and $(\vec{\lambda}^*, \nu^*)$ that satisfy the KKT conditions, then $\vec{x}^*$ will be a primal optimal point, $(\vec{\lambda}^*, \nu^*)$ will be dual optimal. We therefore attempt to find solutions for the KKT conditions. As a first step, show how to simplify the KKT conditions so that they are expressed in terms of only $\vec{x}^*$ and ν^*, i.e. show how $\vec{\lambda}^*$ can be eliminated from these conditions.

(e) Solve for x_i^*, $1 \leq i \leq n$, in terms of ν^* from the simplified KKT conditions derived in the previous subpart.

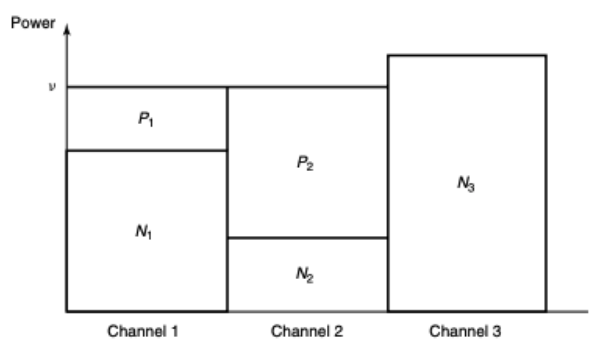

Figure 1: This graphic depicts a solution to the water-filling problem. On the x-axis we have n communication channels and on the y-axis we have the power in each channel. There is a base amount of noise N_i, which for us corresponds to α_i. Water-filling tells us that we should fill each channel until $\frac{1}{\nu^*}$, adding $\frac{1}{\nu^*} - \alpha_i$ power (in this graphic written as P_i), unless α_i already exceeds $\frac{1}{\nu^*}$. One algorithm for achieving this is to allot power to the channel with the least noise until it matches the channel with the second-least noise. Then we fill both simultaneously until they match the level of the channel with the third-least noise. Repeating this process until we run out of power to allot. This distribution of power is akin to filling connected basins with water, hence the name 'water filling'. Figure taken from *Elements of Information Theory* by Cover and Thomas.

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6. Homework Process

With whom did you work on this homework? List the names and SIDs of your group members.

NOTE: If you didn't work with anyone, you can put "none" as your answer.

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